NAVTTC Task Report: Task 02

Course: Artificial Intelligence, Machine Learning & Deep Learning

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Topic: Week 1 — Linux Commands Practice & Verification

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1. Objective

The goal of this task is to practice basic Linux commands and understand how to manage files, directories, and system operations using the terminal.

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2. Practice Tasks

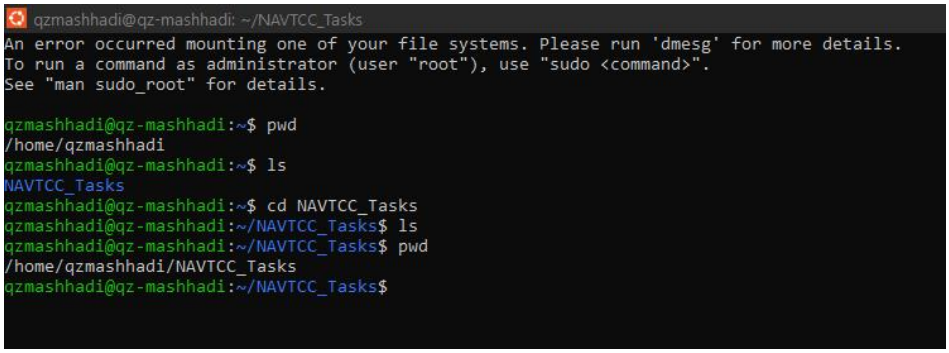
Task A: Basic Navigation Commands

Statement: I have successfully used basic navigation commands to explore directories in Linux.

Commands Used:

- pwd
- ls
- cd

📸 Screenshot:



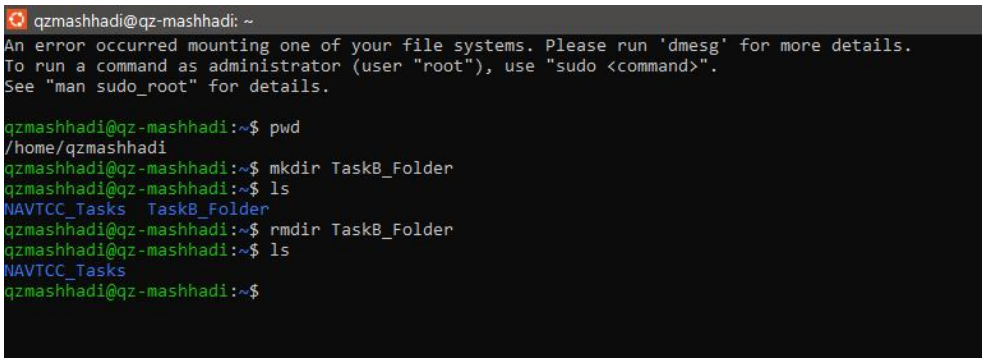
Task B: Directory Management

Statement: I have created and removed directories using Linux commands.

Commands Used:

- mkdir
- rmdir

📸 Screenshot:



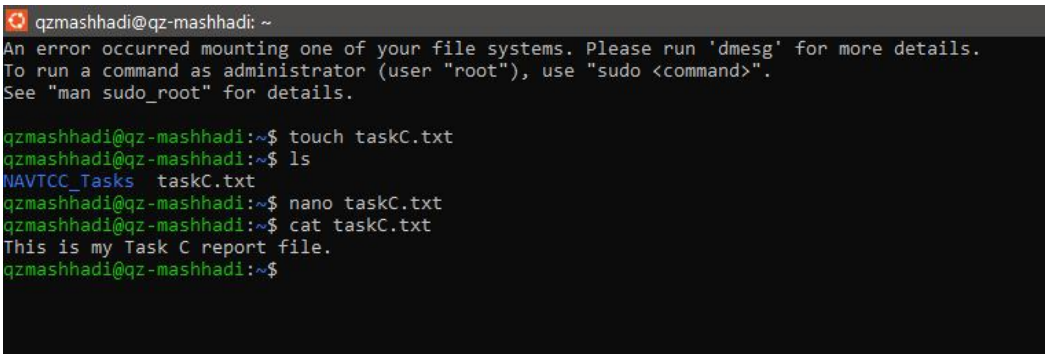
Task C: File Creation and Editing

Statement: I have created and edited files using different Linux commands.

Commands Used:

- touch
- nano
- cat

📸 Screenshot:



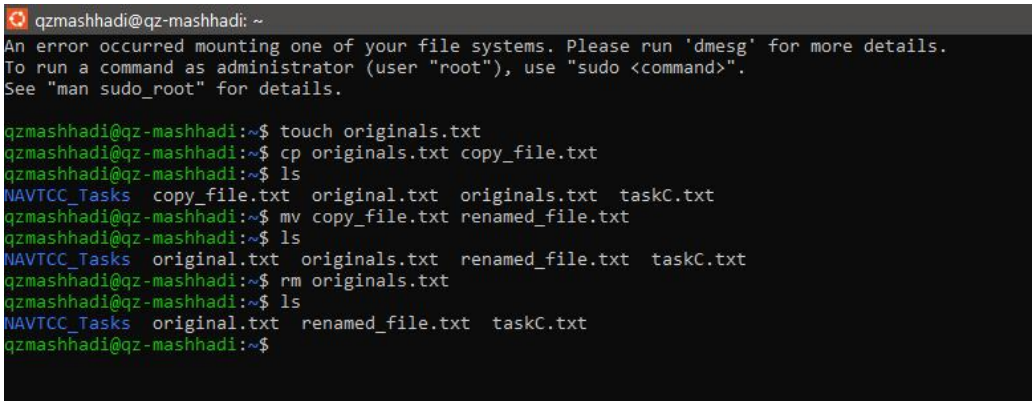
Task D: File Operations

Statement: I have performed file operations like copying, moving, and deleting files.

Commands Used:

- cp
- mv
- rm

📸 Screenshot:



Task E: File Viewing Commands

Statement: I have used different commands to read file contents.

Commands Used:

- cat
- more
- less
- head

📸 Screenshot:

```
qzmashhadi@qz-mashhadi: ~
An error occurred mounting one of your file systems. Please run 'dmesg' for more details.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

qzmashhadi@qz-mashhadi:~$ for i in {1..10}; do echo "Line number $i" >> bigfile.txt; done
qzmashhadi@qz-mashhadi:~$ cat bigfile.txt
Line number 1
Line number 2
Line number 3
Line number 4
Line number 5
Line number 6
Line number 7
Line number 8
Line number 9
Line number 10
qzmashhadi@qz-mashhadi:~$ head -n 5 bigfile.txt
Line number 1
Line number 2
Line number 3
Line number 4
Line number 5
qzmashhadi@qz-mashhadi:~$ more bigfile.txt
Line number 1
Line number 2
Line number 3
Line number 4
Line number 5
Line number 6
Line number 7
Line number 8
Line number 9
Line number 10
qzmashhadi@qz-mashhadi:~$ less bigfile.txt
qzmashhadi@qz-mashhadi:~$
```

Task F: System and Utility Commands

Statement: I have used system-related commands to check processes and manuals.

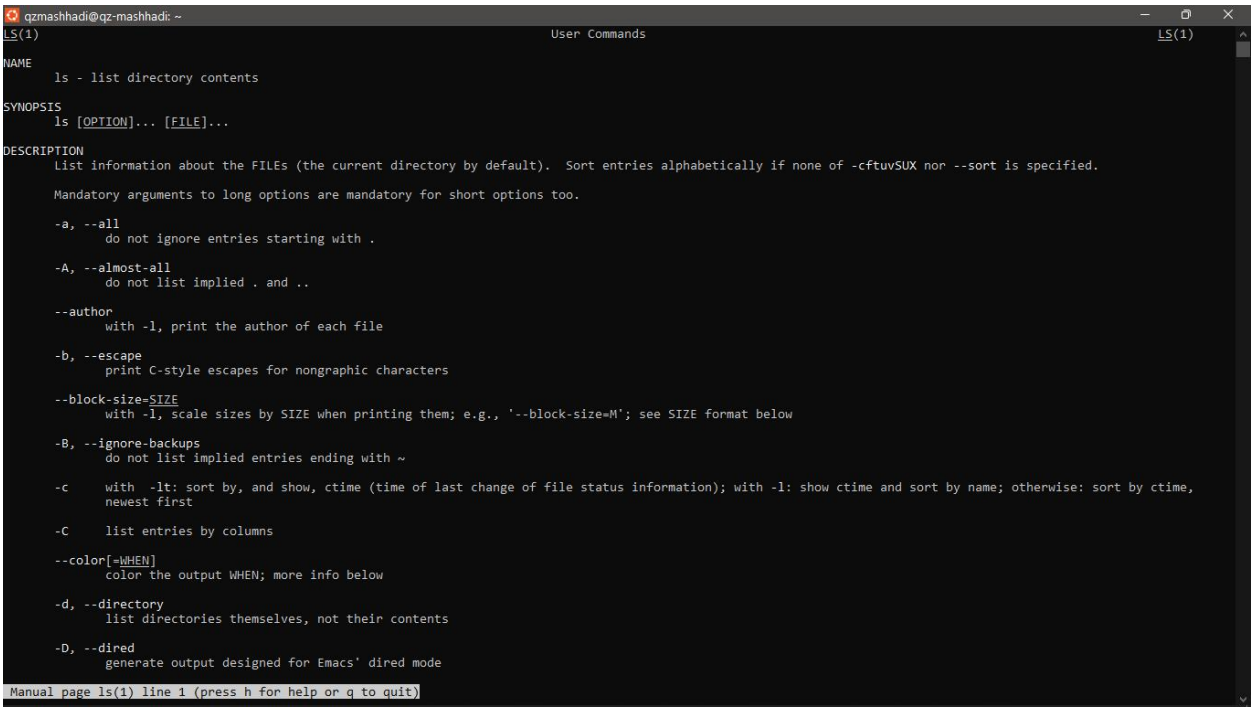
Commands Used:

- ps
- man
- file

📸 Screenshot:

```
qzmashhadi@qz-mashhadi: ~
An error occurred mounting one of your file systems. Please run 'dmesg' for more details.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

qzmashhadi@qz-mashhadi:~$ ps
  PID TTY          TIME CMD
  110 pts/0    00:00:00 bash
  125 pts/0    00:00:00 ps
qzmashhadi@qz-mashhadi:~$ file bigfile.txt
bigfile.txt: ASCII text
qzmashhadi@qz-mashhadi:~$ man ls
qzmashhadi@qz-mashhadi:~$ man ls
qzmashhadi@qz-mashhadi:~$
```



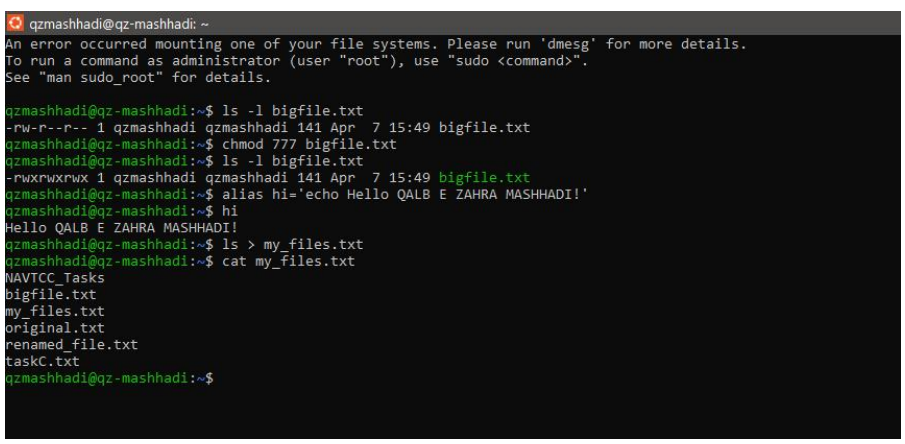
Task G: Advanced Commands

Statement: I have practiced advanced Linux commands for permissions and shortcuts.

Commands Used:

- chmod
- alias
- redirection (>)

📸 Screenshot:



Task H: Administrative Commands

Statement: I have used administrative commands to manage system-level operations.

Commands Used:

- sudo
- shutdown
- reboot
- dpkg

📸 Screenshot:

```
qzmashhadi@qz-mashhadi: ~
An error occurred mounting one of your file systems. Please run 'dmesg' for more details.
sudoTo run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

qzmashhadi@qz-mashhadi:~$ sudo apt update
[sudo] password for qzmashhadi:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1576 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [252 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1875 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [342 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Ign:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1663 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [322 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2894 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [671 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1167 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7520 B]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [500 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7380 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
```

```
qzmashhadi@qz-mashhadi: ~
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1480 B]
Get:42 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:43 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:44 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:45 http://archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:46 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:47 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:49 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:50 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.8 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2757 kB]
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [642 kB]
Get:53 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:54 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6980 B]
Get:57 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:58 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [252 kB]
Fetched 41.1 MB in 4min 53s (140 kB/s)

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
214 packages can be upgraded. Run 'apt list --upgradable' to see them.
qzmashhadi@qz-mashhadi:~$ dpkg --version
Debian 'dpkg' package management program version 1.22.6 (amd64).
This is free software; see the GNU General Public License version 2 or
later for copying conditions. There is NO warranty.
qzmashhadi@qz-mashhadi:~$ shutdown --help
shutdown [OPTIONS...] [TIME] [WALL...]

Shut down the system.

Options:
     --help      Show this help
  -H --halt      Halt the machine
  -P --poweroff  Power-off the machine
  -r --reboot    Reboot the machine
  -h             Equivalent to --poweroff, overridden by --halt
  -k             Don't halt/power-off/reboot, just send warnings
     --no-wall   Don't send wall message before halt/power-off/reboot
  -c             Cancel a pending shutdown
     --show      Show pending shutdown
```

```
qzmashhadi@qz-mashhadi: ~

This is a compatibility interface, please use the more powerful 'systemctl reboot',
'systemctl poweroff', 'systemctl reboot' commands instead.

See the shutdown(8) man page for details.
qzmashhadi@qz-mashhadi:~$ reboot --help
reboot [OPTIONS...] [ARG]

Reboot the system.

Options:
     --help      Show this help
     --halt      Halt the machine
  -p --poweroff  Switch off the machine
     --reboot    Reboot the machine
  -f --force     Force immediate halt/power-off/reboot
  -w --wtmp-only Don't halt/power-off/reboot, just write wtmp record
  -d --no-wtmp   Don't write wtmp record
     --no-wall   Don't send wall message before halt/power-off/reboot

This is a compatibility interface, please use the more powerful 'systemctl reboot' command instead.

See the halt(8) man page for details.
```

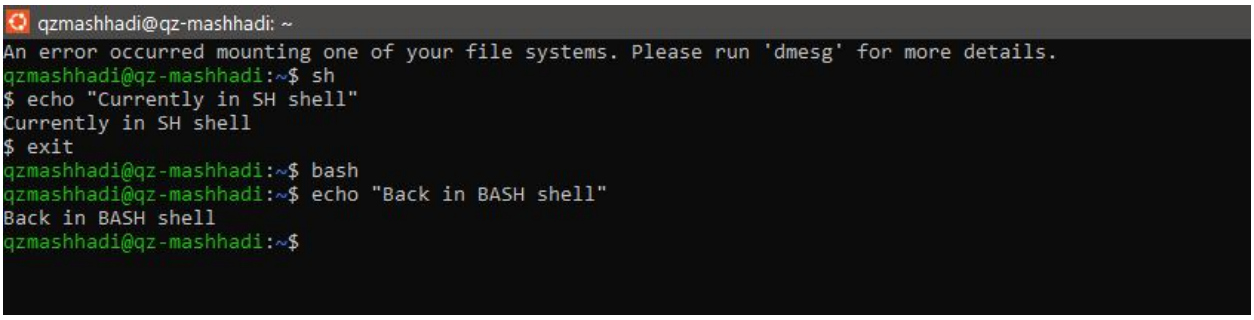
Task I: Shell Commands

Statement: I have used different shell environments in Linux.

Commands Used:

- bash
- sh

📸 Screenshot:



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3. Conclusion

All the basic and important Linux commands have been successfully practiced. I am now able to navigate directories, manage files, and perform system operations using the Linux terminal.